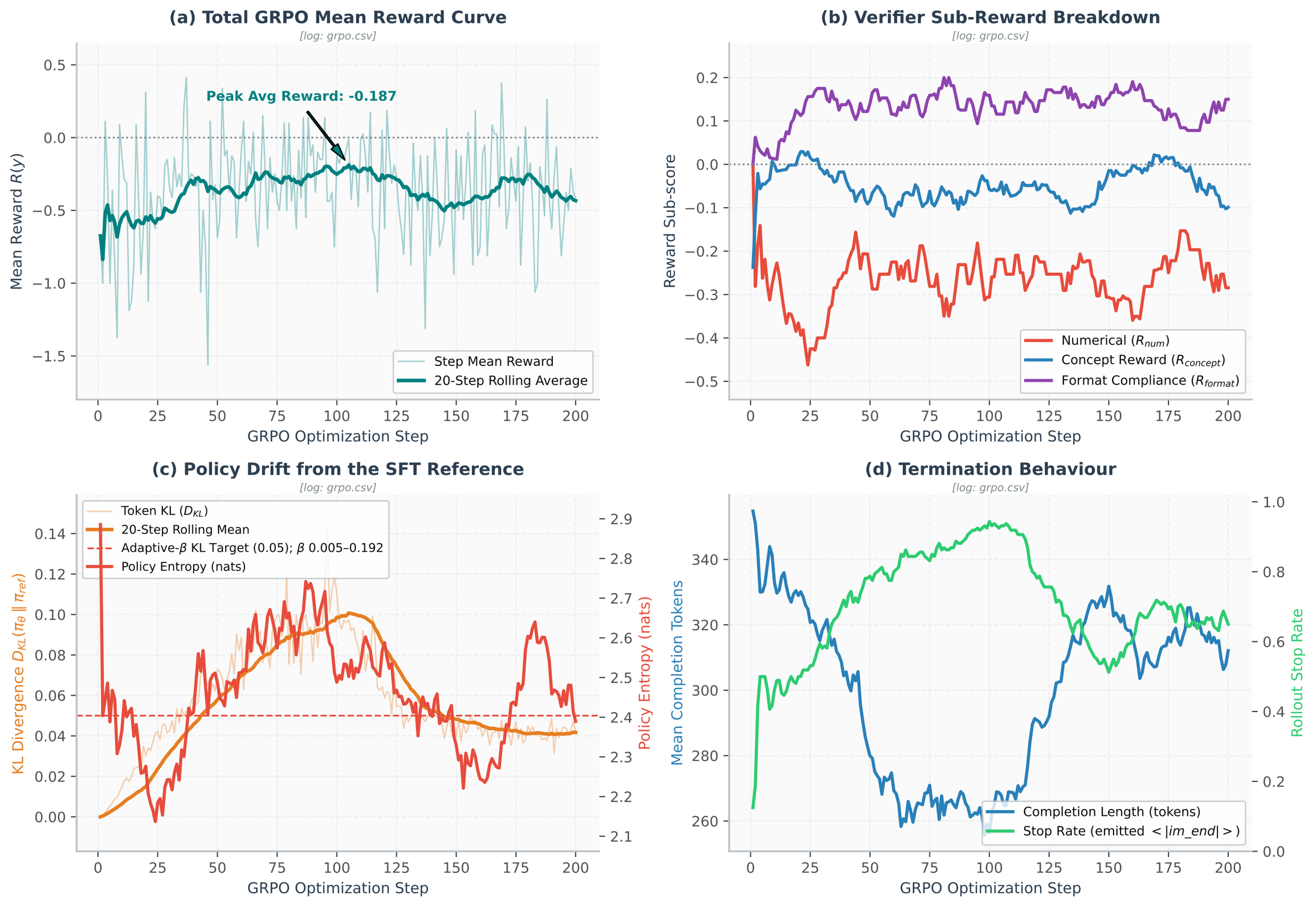

(a) Total GRPO Mean Reward Curve
[log: grpo.csv]
Peak Avg Reward: -0.187
Mean Reward R(y)
GRPO Optimization Step
Step Mean Reward
20-Step Rolling Average
(b) Verifier Sub-Reward Breakdown
[log: grpo.csv]
Reward Sub-score
GRPO Optimization Step
Numerical (R_{num})
Concept Reward ($R_{concept}$)
Format Compliance (R_{format})
(c) Policy Drift from the SFT Reference
[log: grpo.csv]
Token KL (D_{KL})
20-Step Rolling Mean
Adaptive-β KL Target (0.05); β 0.005–0.192
Policy Entropy (nats)
KL Divergence $D_{KL}(\pi_\theta \| \pi_{ref})$
Policy Entropy (nats)
GRPO Optimization Step
(d) Termination Behaviour
[log: grpo.csv]
Mean Completion Tokens
Rollout Stop Rate
GRPO Optimization Step
Completion Length (tokens)
Stop Rate (emitted <|im_end|>)